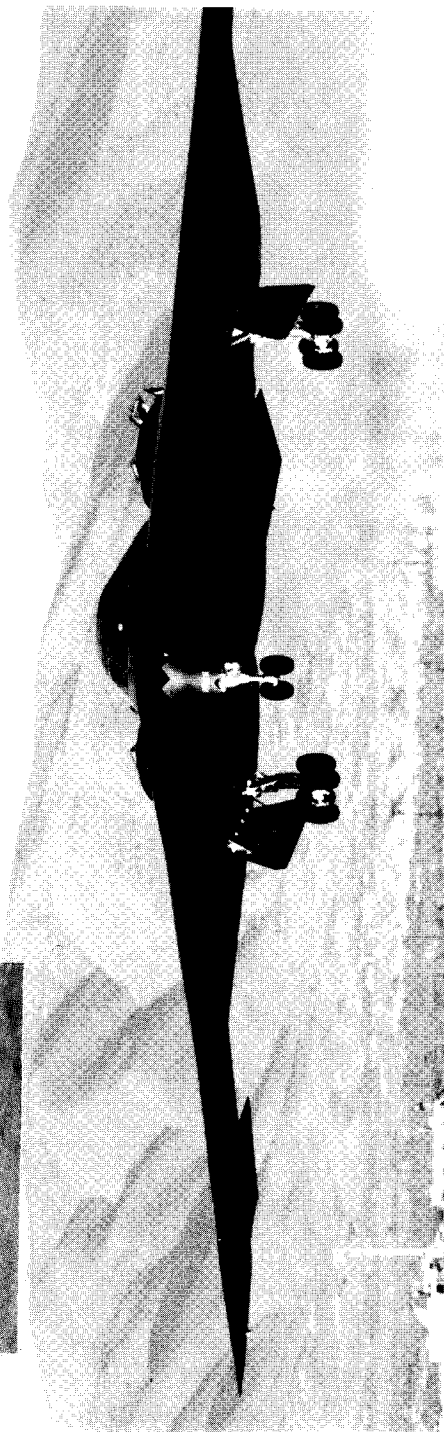
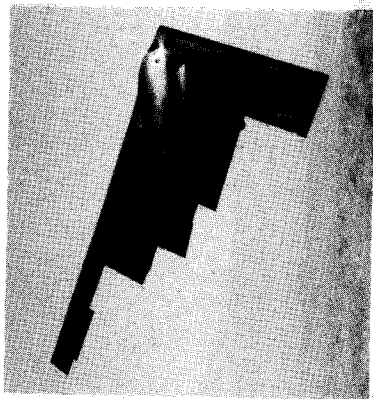




Northrop B-2 Stealth Bomber

SPECIAL CONSIDERATIONS IN CONFIGURATION LAYOUT

8.1 INTRODUCTION

The previous chapter discussed the mechanics of configuration layout. Later chapters will focus on the required provisions for specific internal components, such as the crew station and landing gear. This chapter discusses a number of important intangible considerations, such as aerodynamics, structures, detectability, vulnerability, producibility, and maintainability. All of these are numerically analyzed in later stages of the design process. During configuration layout the designer must consider their impact in a qualitative sense.

8.2 AERODYNAMIC CONSIDERATIONS

The overall arrangement and smoothness of the fuselage can have a major effect upon aerodynamic efficiency. A poorly designed aircraft can have excessive flow separation, transonic drag rise, and supersonic wave drag. Also, a poor wing-fuselage arrangement can cause lift losses or disruption of the desired elliptical lift distribution.

Aerodynamic analysis will be discussed in Chapter 12 and a variety of first-order estimation methods will be presented. During concept layout, the designer must consider the requirements for aerodynamics based upon experience and a "good eye."

Minimization of wetted area is the most powerful aerodynamic consideration for virtually all aircraft. Wetted area directly affects the friction drag. Fuselage wetted area is minimized by tight internal packaging and a low fineness ratio (i.e., a short, fat fuselage). However, excessively tight packaging should be avoided for maintainability considerations. Also, a short, fat fuselage will have a short tail moment arm which increases the required tail areas. The short, fat fuselage will also have high supersonic wave drag.

Another major driver for good aerodynamic design during fuselage layout is the maintenance of smooth longitudinal contours. These can be provided by the use of smooth longitudinal control lines. Generally, longitudinal breaks in contour should follow a radius at least equal to the fuselage diameter at that point.

To prevent separation of the airflow, the aft-fuselage deviation from the freestream direction should not exceed 10–12 deg (Fig. 8.1). However, the air inflow induced by a pusher-propeller will prevent separation despite contour angles of up to 30 deg or more.

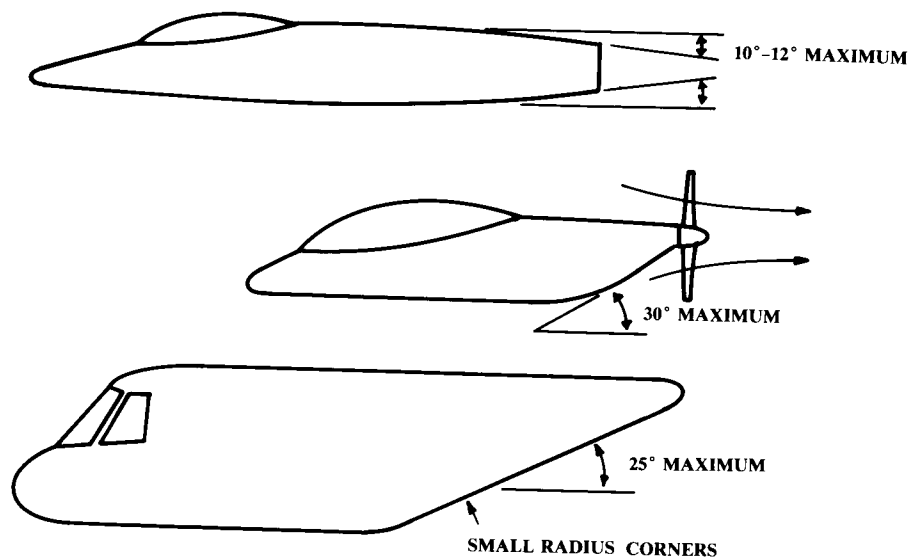


Fig. 8.1 Longitudinal contour guidelines.

A lower-surface upsweep of about 25 deg can be tolerated for a rear-loading transport aircraft provided that the fuselage lower corners are fairly sharp. This causes a vortex-flow pattern that reduces the drag penalty. In general, aft-fuselage upsweep should be minimized as much as possible, especially for high-speed aircraft.

The importance of well-designed wing fillets has already been discussed. Fillets are especially important for low-wing, high-speed aircraft such as jet transports.

These aircraft will frequently have a modified wing-root airfoil to further minimize fuselage interference and shock-induced drag increases. This modification takes the form of an uncambered or even negatively-cambered airfoil set at a high positive angle of incidence. Design of such a wing modification is beyond the scope of this book, but for layout purposes can be approximated by examining the wing of an existing, similar speed-class aircraft.

“Base area” is any unfaired, rearward-facing blunt surface. Base area causes extremely high drag due to the low pressure experienced by the rearward-facing surface (see Chapter 12).

However, a base area between or very near to the jet exhausts may be “filled-in” by the pressure field of the exhaust, partially alleviating the drag penalty. The T-38 has such a base area between its nozzles. A base area fill-in effect is difficult to predict.

The aerodynamic interaction between different components should be visualized in designing the aircraft. For example, a canard should not be located such that its wake might enter the engine inlets at any possible angle of attack. Wake ingestion can stall or even destroy a jet engine.

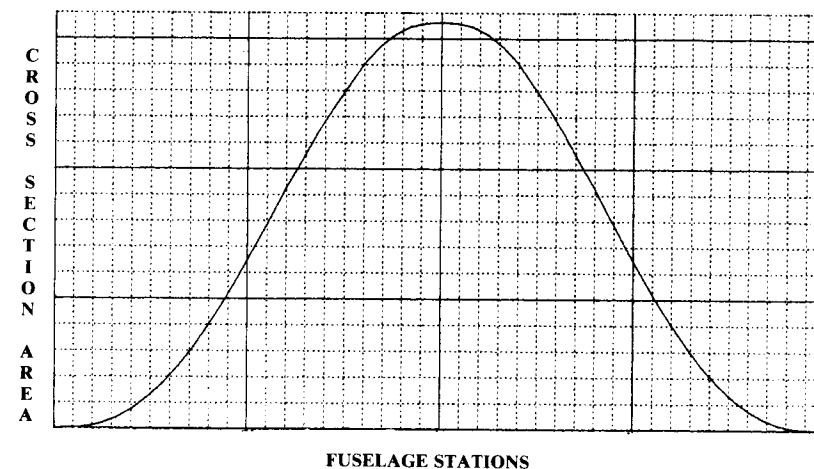


Fig. 8.2 Sears-Haack volume distribution.

If an aircraft’s forebody has sharp lower corners, a separated vortex can be expected at high angle of attack. This could also be ingested by the inlets, with bad results. Also, such a vortex could unpredictably affect the wing or tail surfaces.

For supersonic aircraft, the greatest aerodynamic impact upon the configuration layout results from the desire to minimize supersonic wave drag, a pressure drag due to the formation of shocks. This is analytically related to the longitudinal change in the aircraft’s total cross-sectional area. In fact, wave drag is calculated using the second derivative (i.e., curvature) of the volume-distribution plot as shown in Fig. 7.36.

Thus, a “good” volume distribution from a wave-drag viewpoint has the required total internal volume distributed longitudinally in a fashion that minimizes curvature in the volume-distribution plot. Several mathematical solutions to this problem have been found for simple bodies-of-revolution, with the “Sears-Haack” body (Fig. 8.2—see Ref. 16) having the lowest wave drag.

If an aircraft could be designed with a volume plot shaped like the Sears-Haack volume distribution it would have the minimum wave drag at Mach 1.0 for a given length and total internal volume. (What happens at higher Mach numbers is discussed in Chapter 12, but for initial layout purposes the minimization of wave drag at Mach 1.0 is a suitable goal in most cases.)

However, it is usually impossible to exactly or even approximately match the Sears-Haack shape for a real aircraft. Fortunately, major drag reductions can be obtained simply by smoothing the volume distribution shape.

As shown in Fig. 8.3, the main contributors to the cross-sectional area are the wing and the fuselage. A typical fuselage with a trapezoidal wing will have an irregularly-shaped volume distribution with the maximum cross-sectional area located near the center of the wing. By “squeezing” the

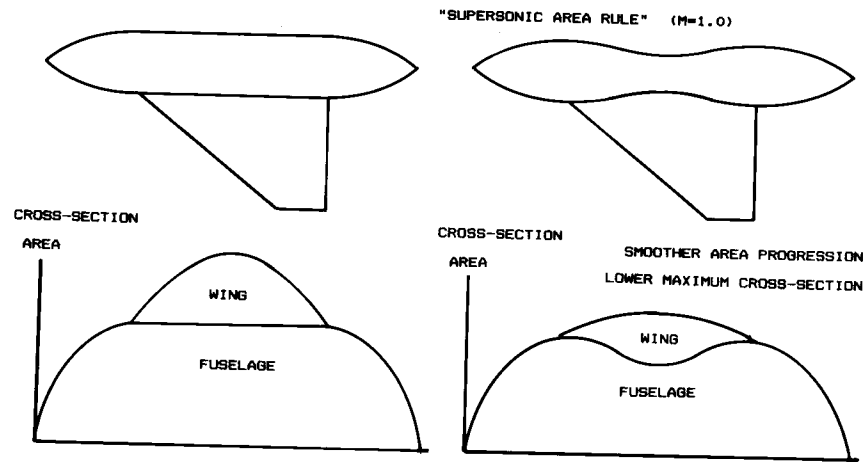


Fig. 8.3 Design for low wave drag.

fuselage at that point, the volume-distribution shape can be smoothed and the maximum cross-sectional area reduced.

This design technique, developed by R. Whitcomb of the NACA (Ref. 17), is referred to as “area-ruling” or “coke-bottling” and can reduce the wave drag by as much as 50%. Note that the volume removed at the center of the fuselage must be provided elsewhere, either by lengthening the fuselage or by increasing its cross-sectional area in other places.

While area-ruling was developed for minimization of supersonic drag, there is reason to believe that even low-speed aircraft can benefit from it to some extent. The airflow over the wing tends to separate toward the trailing edge. If an aircraft is designed such that the fuselage is increasing in cross-sectional area towards the wing trailing edge, this may “push” air onto the wing, thus reducing the tendency to separate. The Wittman Tailwind, which is remarkably efficient, uses this approach.

8.3 STRUCTURAL CONSIDERATIONS

In most larger companies, the configuration designer is not ultimately responsible for the structural arrangement of the aircraft. That is the responsibility of the structural design group. However, a good configuration designer will consider the structural impacts of the general arrangement of the aircraft, and will in fact have at least an initial idea as to a workable structural arrangement.

The primary concern in the development of a good structural arrangement is the provision of efficient “load paths”—the structural elements by which opposing forces are connected. The primary forces to be resolved are the lift of the wing and the opposing weight of the major parts of the aircraft, such as the engines and payload. The size and weight of the structural members will be minimized by locating these opposing forces near to each other.

Carried to the extreme, this leads to the Flying Wing concept. In a flying wing the lift and weight forces can be located at virtually the same place. In the ideal case, the weight of the aircraft would be distributed along the span of the wing exactly as the lift is distributed (Fig. 8.4). This is referred to as “spanloading,” and eliminates the need for a heavy wing structure to carry the weight of the fuselage to the opposing lift force exerted by the wing. The structure can then be sized by lesser requirements such as the landing-gear loads.

While ideal span-loading is rarely possible, the span-loading concept can be applied to more-conventional aircraft by spreading some of the heavy items such as engines out along the wing. This will yield noticeable weight savings, but must be balanced against the possible drag increase.

If the opposing lift and weight forces cannot be located at the same place, then some structural path will be required to carry the load. The weight of structural members can be reduced by providing the shortest, straightest load path possible.

Figure 8.5 illustrates a structural arrangement for a small fighter. The major fuselage loads are carried to the wing by “longerons,” which are typically I- or H-shaped extrusions running fore and aft and attached to the

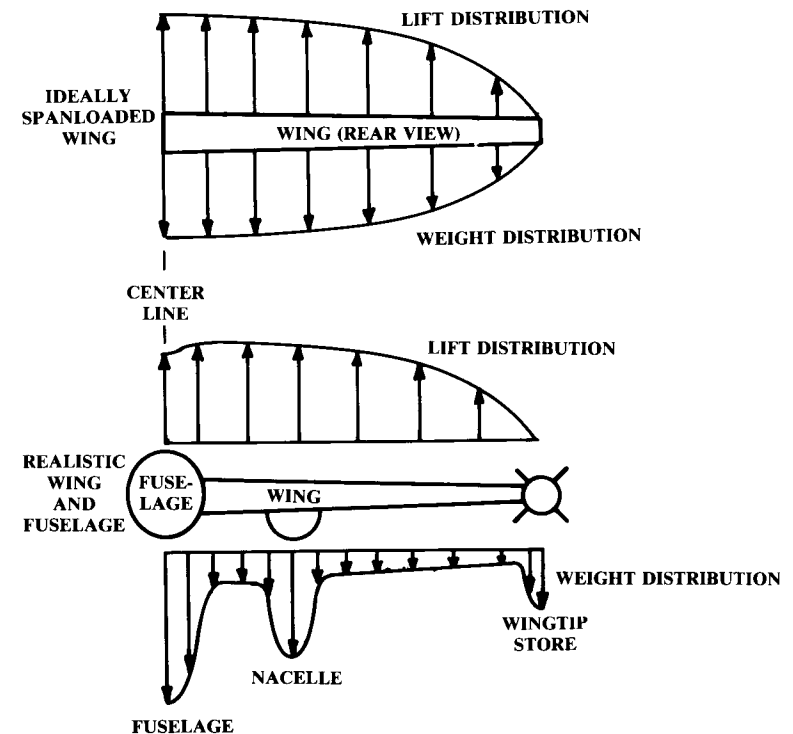


Fig. 8.4 Spanloading for weight reduction.

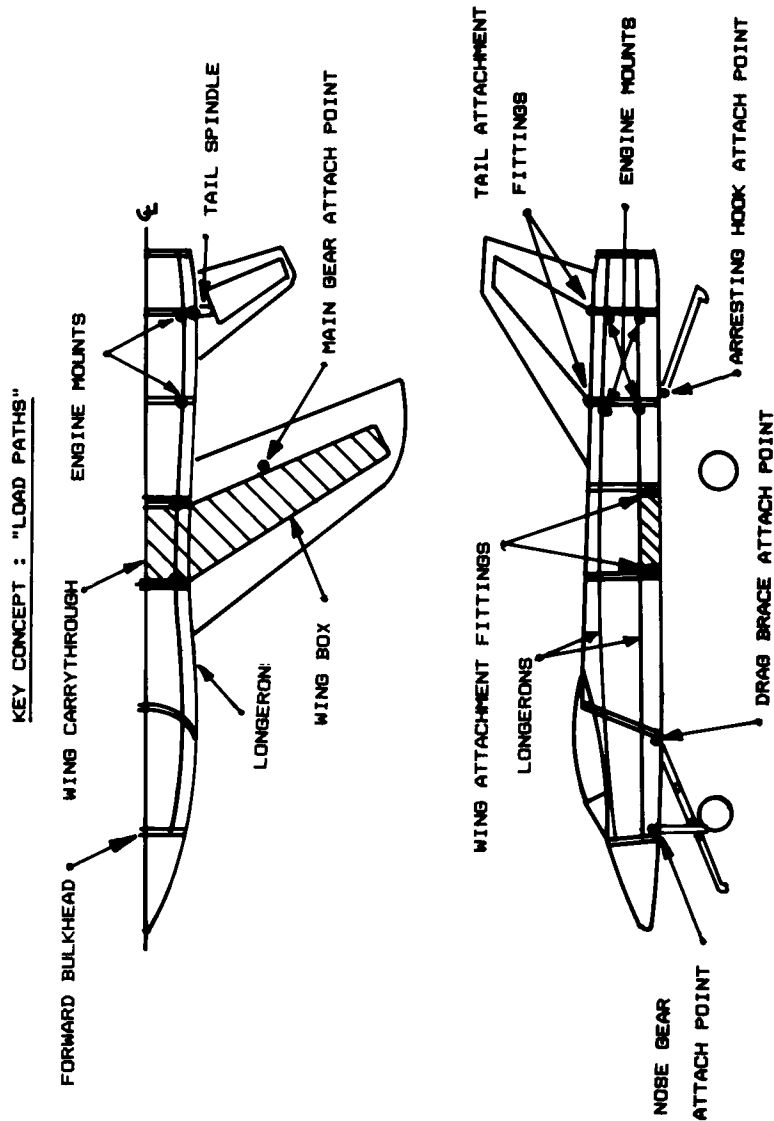


Fig. 8.5 Structural arrangement.

skin. Longerons are heavy, and their weight should be minimized by designing the aircraft so that they are as straight as possible.

For example, the lower longerons in Fig. 8.5 are high enough that they pass over the wing-carrythrough box. Had the longerons been placed lower, they would have required a kink to pass over the box.

On the other hand, the purpose of the longeron is to prevent fuselage bending. This implies that the lightest longeron structure occurs when the upper and lower longerons are as far apart vertically as possible. In Fig. 8.6 the longerons are farther apart, but this requires a kink to pass over the box. Only a trade study can ultimately determine which approach is lighter for any particular aircraft.

In some designs similar to Fig. 8.5 the lower longerons are placed near the bottom of the aircraft. A kink over the wing box is avoided by passing the longeron under or through the wing box. This minimizes weight but complicates both fabrication and repair of the aircraft.

For aircraft such as transports, which have fewer cutouts and concentrated loads than a fighter, the fuselage will be constructed with a large number of "stringers" which are distributed around the circumference of the fuselage (Fig. 8.7). Weight is minimized when the stringers are all straight and uninterrupted.

Another major structural element used to carry fuselage bending loads is the "keelson." This is like the keel on a boat, and is a large beam placed at the bottom of the fuselage as shown in Fig. 8.7. A keelson is frequently used to carry the fuselage bending loads through the portion of the lower fuselage which is cut up by the wheel wells.

As the wing provides the lift force, load-path distances can be reduced by locating the heavy weight items as near to the wing as possible. Similarly, weight can be reduced by locating structural cutouts away from the wing. Required structural cutouts include the cockpit area and a variety of doors (passenger, weapons bay, landing gear, engine access, etc.).

An especially poor arrangement (seen on some older fighter aircraft) has the main landing gear retracting into the wing-box area, which requires a large cutout where the loads are the greatest.

When possible, structural cutouts should be avoided altogether. For example, a jet engine that is buried in the fuselage requires a cutout for the inlet, a cutout for the exhaust, and in most cases another cutout for removal

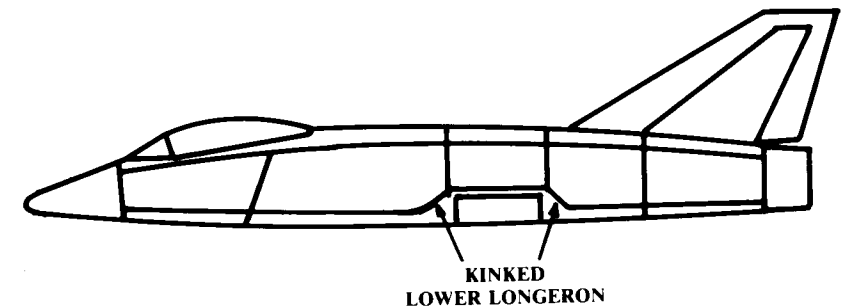


Fig. 8.6 Kinked lower longeron.

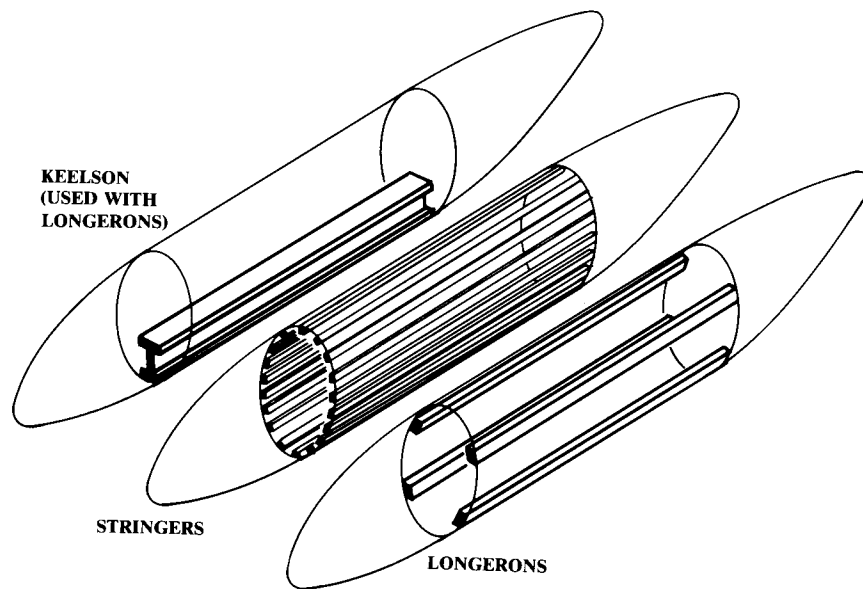


Fig. 8.7 Structural concepts for fuselage loads.

of the engine. The resulting weight penalty compared to a podded engine must be balanced against the reduced drag of a buried engine installation.

Figure 8.5 illustrates another important concept in structural arrangement. Large concentrated loads such as the wing and landing gear attachments must be carried by a strong, heavy structural member such as a major fuselage bulkhead. The number of such heavy bulkheads can be minimized by arranging the aircraft so that the bulkheads each carry a number of concentrated loads, rather than requiring a separate bulkhead for each concentrated load.

In Fig. 8.5 the two bulkheads in the aft fuselage carry the loads for the engines, tails, and arresting hook. Had the tails and engine been located without this in mind, the structural designer would have had to provide four or five heavy bulkheads rather than the two shown.

The lift force on the wing produces a tremendous bending moment where the wing attaches to the fuselage. The means by which this bending moment is carried across the fuselage is a key parameter in the structural arrangement, and will greatly influence both the structural weight and the aerodynamic drag of the aircraft.

Figure 8.8 illustrates the four major types of wing carrythrough structure. The “box carrythrough” is virtually standard for high-speed transports and general-aviation aircraft. The box carrythrough simply continues the wing box through the fuselage. The fuselage itself is not subjected to any of the bending moment of the wing, which minimizes fuselage weight.

However, the box carrythrough occupies a substantial amount of fuselage volume, and tends to add cross-sectional area at the worst possible place for

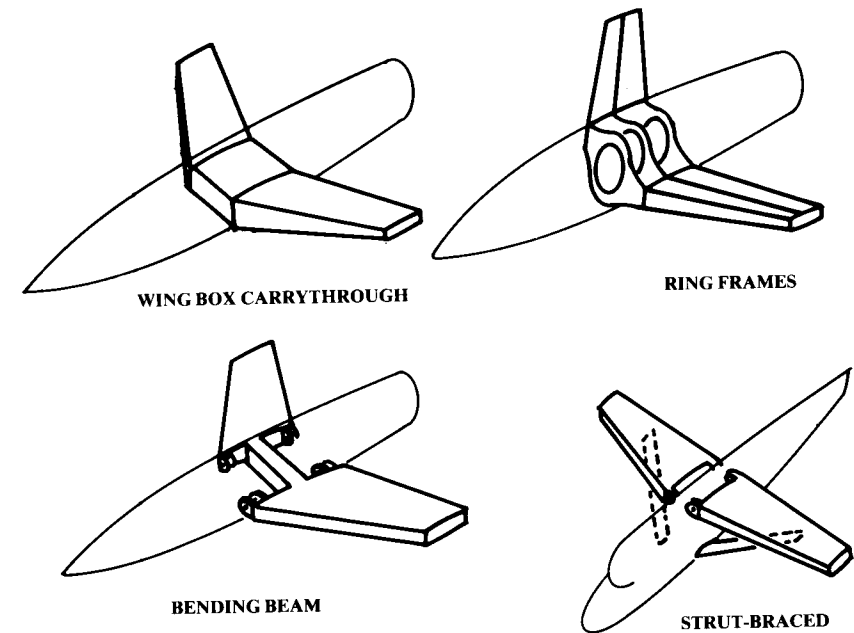


Fig. 8.8 Wing carrythrough structure.

wave drag, as discussed above. Also, the box carrythrough interferes with the longeron load-paths.

The “ring-frame” approach relies upon large, heavy bulkheads to carry the bending moment through the fuselage. The wing panels are attached to fittings on the side of these fuselage bulkheads. While this approach is usually heavier from a structural viewpoint, the resulting drag reduction at high speeds has led to the use of this approach for most modern fighters.

The “bending beam” carrythrough can be viewed as a compromise between these two approaches. Like the ring-frame approach, the wing panels are attached to the side of the fuselage to carry the lift forces. However, the bending moment is carried through the fuselage by one or several beams that connect the two wing panels. This approach has less of a fuselage volume increase than does the box-carrythrough approach.

Many light aircraft and slower transport aircraft use an external strut to carry the bending moments. While this approach is probably the lightest of all, it obviously has a substantial drag penalty at higher speeds.

Aircraft wings usually have the front spar at about 20–30% of the chord back from the leading edge. The rear spar is usually at about the 60–75% chord location. Additional spars may be located between the front and rear spars forming a “multispar” structure. Multispar structure is typical for large or high-speed aircraft.

If the wing skin over the spars is an integral part of the wing structure, a “wing box” is formed which in most cases provides the minimum weight.

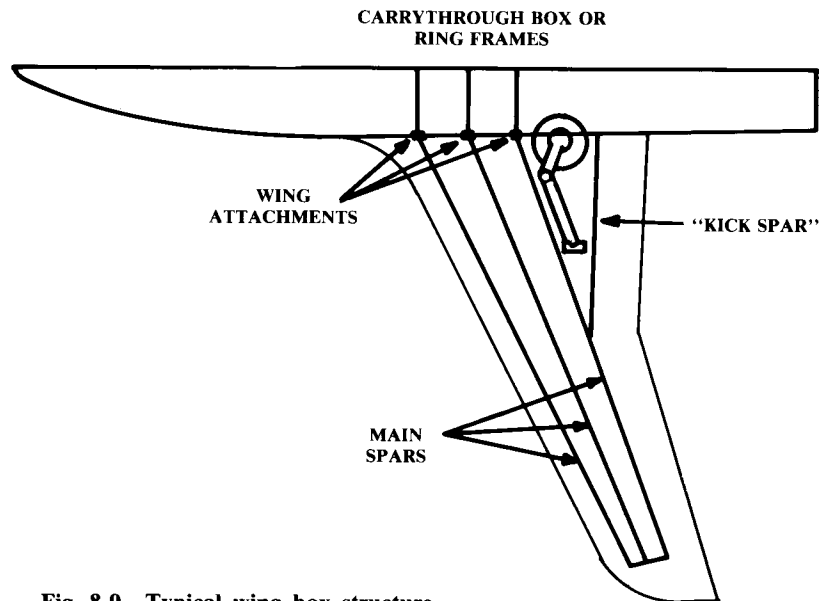


Fig. 8.9 Typical wing box structure.

Aircraft with the landing gear in the wing will usually have the gear located aft of the wing box, with a single trailing-edge spar behind the gear to carry the flap loads, as shown in Fig. 8.9.

Ribs carry the loads from the control surfaces, store stations, and landing gear to the spars and skins. A multispar wing box will have comparatively few ribs, located only where major loads occur.

Another form of wing structure, the “multirib” or “stringer panel” box, has only two spars, plus a large number of spanwise stringers attached to the wing skins. Numerous ribs are used to maintain the shape of the box under bending.

Variable sweep and folding capability add considerably to the wing structural weight. On the other hand, use of a delta wing will reduce the structural weight. These are further discussed in Chapter 15.

First-order structural sizing will be discussed in Chapter 14. For initial layout purposes the designer must guess at the amount of clearance required for structure around the internal components. A good designer with a “calibrated eyeball” can prevent a lot of lost effort, for the aircraft may require substantial redesign if later structural analysis determines that more room is required for the structural members.

A large airliner will typically require about 4 in. of clearance from the inner wall of the passenger compartment to the outer skin (“moldline”). The structure of a conventional fighter fuselage will typically require about 2 in. of offset from the moldline for internal components. For a small general aviation aircraft, 1 in. clearance or less may be acceptable.

The type of internal component will affect the required clearance. A jet engine contained within an aluminum or composite fuselage will require

perhaps an additional inch of clearance to allow for a heat shield. The heat shield may be constructed of titanium, steel, or a heat-proof matting. On the other hand, an “integral” fuel tank in which the existing structure is simply sealed and filled with fuel will require no clearance other than the thickness of the skin.

There is no easy formula for the estimation of structural clearance. The designer must use judgement acquired through experience. The best way to gain this judgement other than actual design experience is by looking at existing designs.

8.4 RADAR DETECTABILITY

Ever since the dawn of military aviation attempts have been made to reduce the detectability of aircraft. During World War I, the only “sensor” in use was the human eyeball. Camouflage paint in mottled patterns was used on both sides to reduce the chance of detection.

Radar (acronym for Radio Detection And Ranging), the primary sensor used against aircraft today, consists of a transmitter antenna that broadcasts a directed beam of electromagnetic radio waves and a receiver antenna which picks up the faint radio waves that bounce off objects “illuminated” by the radio beam. Usually the transmitter and receiver antennas are collocated (“monostatic radar”), although some systems have them in different locations (“bistatic radar”).

Detectability to radar has been a concern since radar was first used in World War II. “Chaff” was the first radar “stealth” technology. Chaff, also called “window,” consists of bits of metal foil or metallized fibers dropped by an aircraft to create many radar echos that hide its actual echo return. Chaff is still useful against less-sophisticated radars.

Chaff obscures the actual location of the aircraft, but does not allow the aircraft to pass unnoticed. To avoid detection, the aircraft must return such a low amount of the transmitted radio beam that the receiver antenna cannot distinguish between it and the background radio static.

The extent to which an object returns electromagnetic energy is the object’s “Radar Cross Section” (RCS). RCS is usually measured in square meters or in decibel square meters, with “zero dBsm” equal to ten to the zero power, or one square meter. “Twenty dBsm” equals ten to the second power, or 100 square meters. Because radar signal strength is an inverse function of the fourth power of the distance to the target, it takes a very substantial reduction in RCS to obtain a meaningful operational benefit.

Actually, the RCS of an aircraft is not a single number. The RCS is different for each “look-angle” (i.e., direction from the threat radar). Also, the RCS varies depending upon the frequency and polarization of the threat radar (see Ref. 21). The following comments relate to typical threat radars seen by military aircraft.

There are many electromagnetic phenomena that contribute to the RCS of an aircraft. These require different design approaches for RCS reduction, and can produce conflicting design requirements. Figure 8.10 illustrates the major RCS contributors for a typical, untreated fighter aircraft.

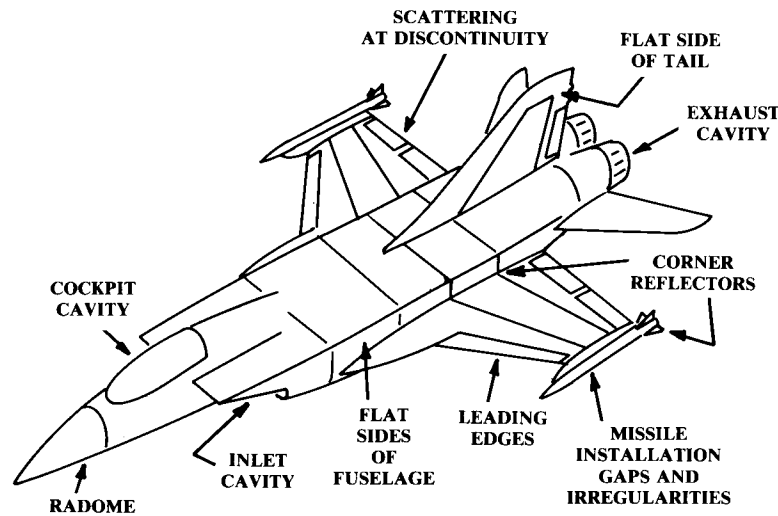


Fig. 8.10 Major RCS contributors.

One of the largest contributions to airframe RCS occurs any time a relatively flat surface of the aircraft is perpendicular to the incoming radar beam. Imagine shining a flashlight at a shiny aircraft in a dark hanger. Any spots where the beam is reflected directly back at you will have an enormous RCS contribution.

Typically this “specular return” occurs on the flat sides of the aircraft fuselage and along an upright vertical tail (when the radar is abeam the aircraft). To prevent these RCS “spikes,” the designer may slope the fuselage sides, angle the vertical tails, and so on, so that there are no flat surfaces presented towards the radar (Fig. 8.11).

Note that this RCS reduction approach assumes that the designer knows where the threat radar will be located relative to the aircraft. This information is usually provided by the operations-analysis department as a design driver. Also, this assumes a monostatic radar.

Another area of the aircraft which can present a perpendicular bounce for the radar is the round leading edge of the wing and tail surfaces. If the aircraft is primarily designed for low detectability by a nose-on threat radar, the wings and tails can be highly swept to reduce their contribution to RCS. Note that this and many other approaches to reducing the RCS will produce a penalty in aerodynamic efficiency.

It is also important to avoid any “corner reflectors,” i.e., intersecting surfaces that form approximately a right angle, as shown in Fig. 8.10 at the wing-fuselage junction.

Another contributor to airframe RCS occurs due to the electromagnetic currents that build up on the skin when illuminated by a radar. These currents flow across the skin until they hit a discontinuity such as at a sharp

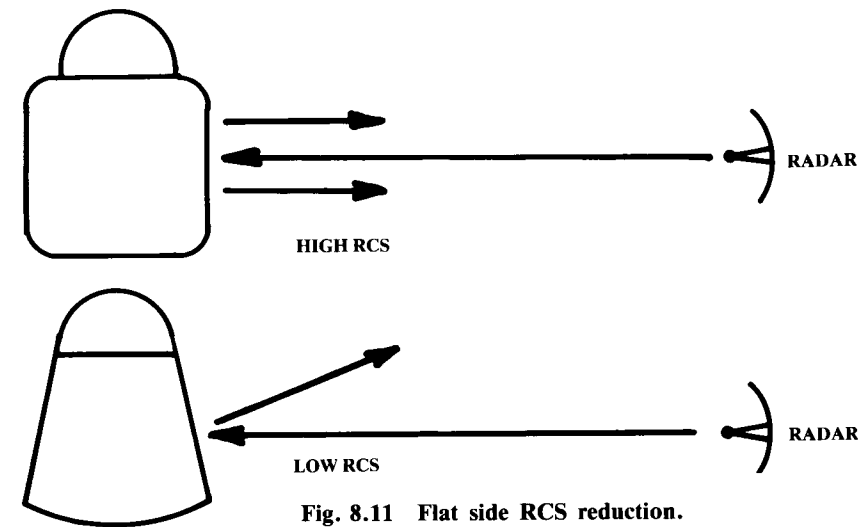


Fig. 8.11 Flat side RCS reduction.

trailing edge, a wing tip, a control surface, or a crack around a removable panel or door. At a discontinuity, the currents “scatter,” or radiate electromagnetic energy, some of which is transmitted back to the radar (Fig. 8.12).

This effect is much lower in intensity than the specular return, but is still sufficient for detection. The effect is strongest when the discontinuity is straight and perpendicular to the radar beam. Thus, the discontinuities such as at the wing and tail trailing edges can be swept to minimize the detectability from the front. Carried to the extreme, this leads to diamond- or sawtooth-shaped edges on every door, access plate, and other discontinuity on the aircraft, as seen on the B-2 and F-117.

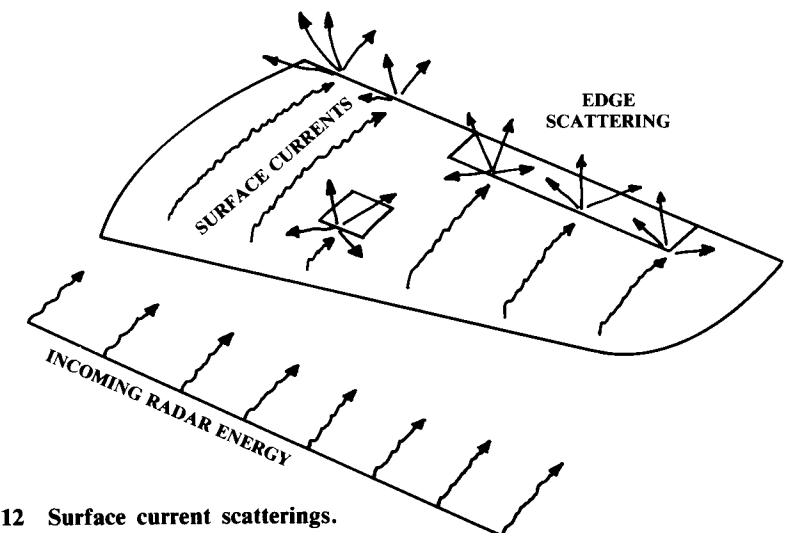


Fig. 8.12 Surface current scatterings.

First-generation stealth designs such as the Lockheed F-117 and the never-constructed North American Rockwell "Surprise Fighter" relied upon faceted shaping in which the aircraft shape is constructed of interlocking flat triangles and trapezoids. This has advantages in ease of construction and signature analysis, but offers a large number of sharp edges to create diffraction returns, and so is no longer in favor (Ref. 92).

Current stealth design begins by configuring the aircraft such that all "big" returns, such as from perpendicular bounces, are "aimed" in just a few directions. For example, if the leading edges of the wings and tails are all straight and set at the same angle, there would be a huge radar return from that angle direction, but little return from other directions. This would presumably offer a small probability that the aircraft and threat radar would be mutually oriented in exactly the angle of high return, and the aircraft would be undetectable from all other angles.

It is also common practice to "aim" the wing leading-edge return in the same direction as the edge diffraction return from the trailing edge. This is done either by using identical sweep angles at the leading and trailing edges (thus, a wing with no taper, as on the B-2), or by aligning the left wing leading edge at the same sweep angle as the right wing trailing edge (and vice versa). This creates a diamond wing as on the F-23 and an early Mikoyan research aircraft.

Once all wing and tail returns are "aimed" in the same direction, the returns from doors, access panels, and other discontinuities can be "aimed" in the same direction by alignment of their edges. This is clearly seen on the B-2, where virtually every feature on the aircraft, including weapons, bay doors, gear doors, inlets, nozzles, and access panels, is constructed using only lines which are parallel to a wing leading edge.

This design approach leads to an aircraft planform composed entirely of straight, highly swept lines, much like the first-generation stealth designs. However, the desire to eliminate the edge diffractions caused by the facets of first-generation stealth now produces designs in which cross-sectional shapes are smooth, not sharp-edged. The steep angles on the fuselage sides as shown in Fig. 8.11 are employed to prevent broadside perpendicular bounce returns, but these angled sides flow smoothly over the top and bottom of the fuselage. Such shaping can be seen on the B-2, F-22, and F-23.

RCS can also be reduced simply by eliminating parts of the aircraft. A horizontal tail that isn't there cannot contribute to the radar return! Modern computerized flight controls combined with the use of vectored-thrust engines can solve many of the difficulties of the tailless configuration.

Similarly, RCS can be reduced if the nacelles can be eliminated through the use of buried engines, or better yet, by eliminating the entire fuselage through the use of the flying-wing concept. This approach is used in the Northrop B-2.

In addition to reshaping the aircraft, detectability can be reduced through the use of skin materials that absorb radar energy. Such materials, called "radar absorbing materials" (RAM), are typically composites such as fiberglass embedded with carbon or ferrite particles.

These particles are heated by the radar electromagnetic waves, thus absorbing some of the energy. This will reduce (not eliminate!) the radar

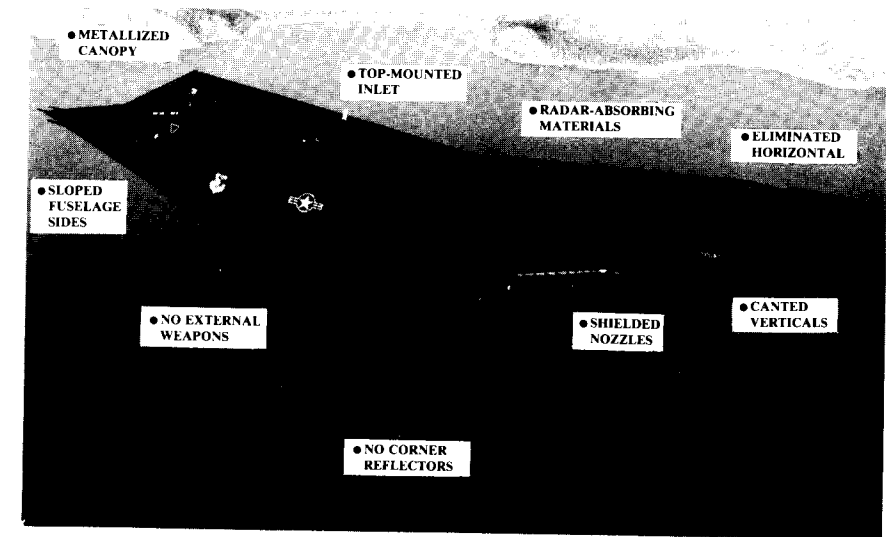


Fig. 8.13 Detectability reduction approaches.

return due to perpendicular bounce, and can also reduce the surface currents and thus reduce the RCS due to scattering at sharp edges.

As there are many types of RAM and similar treatments, no quick estimate for the weight impact of their use can be provided here. However, one can probably assume that such use will reduce or eliminate any weight savings otherwise assumed for the use of composite materials.

For most existing aircraft, the airframe is not the largest contributor to RCS, especially nose-on. A conventional radome, covering the aircraft's own radar, is transparent to radar for obvious reasons. Therefore, it is also transparent to the threat radar, allowing the threat radar's beam to bounce off the forward bulkhead and electronic equipment within the radome.

Even worse, the aircraft's own radar antenna, when illuminated by a threat radar, can produce a radar magnification effect much like a cat's eye. These effects can be reduced with a "bandpass" radome, which is transparent to only one radar frequency (that of the aircraft's radar).

Other huge contributors to the RCS for a conventional aircraft are the inlet and exhaust cavities. Radar energy gets into these cavities, bounces off the engine parts, and sprays back out the cavity towards the threat radar. Also, these cavities represent additional surface discontinuities.

The best solution for reducing these RCS contributions is to hide them from the expected threat locations. For example, inlets can be hidden from ground-based radars by locating them on top of the aircraft (Fig. 8.13). Exhausts can be hidden through the use of two-dimensional nozzles.

Cockpits provide a radar return for a similar reason. The radar energy enters the cockpit, bounces around off the equipment inside, and then reradiates back outside. One solution for this is to thinly coat the canopy with some conductive metal such as gold, causing the canopy to reflect the radar energy away.

Finally, the aircraft's weapons can have a major impact on RCS. Missiles and bombs have fins that form natural corner reflectors. The carriage and release mechanisms have numerous corner reflectors, cavities, and surface discontinuities. Gun ports present yet another kind of cavity. The only real solution for these problems is to put all the weapons inside, behind closed doors. However, the weight, volume, and complexity penalties of this approach must be carefully considered.

Electronic countermeasures (ECM)—devices to trick the threat radar—usually consist of some sort of radar receiver that picks up the threat radar emissions, and some sort of transmit antenna to send a deceiving signal back to the threat radar. The many techniques for tricking radar (and ECM) go beyond the scope of this book. However, designers should be aware that there is a tradeoff between the aircraft's RCS level and the required amount of ECM.

8.5 INFRARED DETECTABILITY

Infrared (IR) detectability also concerns the aircraft designer. Many short-range air-to-air and ground-to-air missiles rely upon IR seekers. Modern IR sensors are sensitive enough to detect not only the radiation emitted by the engine exhaust and hot parts, but also that emitted by the whole aircraft skin due to aerodynamic heating at transonic and supersonic speeds. Also, sensors can detect the solar IR radiation that reflects off the skin and cockpit transparencies (windows).

Of several approaches for reduction of IR detectability, one of the most potent reduces engine exhaust temperatures through the use of a high-bypass-ratio engine. This reduces both exhaust and hot-part temperatures. However, depending upon such an engine for IR reduction may result in selecting one that is less than optimal for aircraft sizing, which increases aircraft weight and cost.

Emissions from the exposed engine hot-parts (primarily the inside of the nozzle) can be reduced by cooling them with air bled off the engine compressor. This will also increase fuel consumption slightly. Another approach hides the nozzles from the expected location of the threat IR sensor. For example, the H-tails of the A-10 hide the nozzles from some angles. Unfortunately, the worst-case threat location is from the rear, and it is difficult to shield the nozzles from that direction!

Plume emissions are reduced by quickly mixing the exhaust with the outside air. As mentioned, a high-bypass engine is the best way of accomplishing this. Mixing can also be enhanced by the use of a wide, thin nozzle rather than a circular one. Another technique is to angle the exhaust upward or downward relative to the freestream. This will have an obvious thrust penalty, however.

Sun glint in the IR frequencies can be somewhat reduced by the use of special paints that have low IR reflectivity. Cockpit transparencies (which can't be painted!) can be shaped with all flat sides to prevent continuous tracking by an IR sensor.

Emissions due to aerodynamic heat are best controlled by slowing the aircraft down.

IR missiles can sometimes be tricked by throwing out a flare which burns to produce approximately the same IR frequencies as the aircraft. However, modern IR seekers are getting better at identifying which hot source is the actual aircraft.

IR fundamentals are more thoroughly discussed in Ref. 18.

8.6 VISUAL DETECTABILITY

The human eyeball is still a potent aircraft-detection sensor. On a clear day, an aircraft or its contrail may be spotted visually before detection by the on-board radar of a typical fighter. Also, fighter aircraft usually have radar only in front, which leaves the eyeball as the primary detector for spotting threat aircraft which are abeam or above.

Visual detection depends upon the size of aircraft and its color and intensity contrast with the background. In simulated combat, pilots of the small F-5 can frequently spot the much-larger F-15s well before the F-5s are seen. However, aircraft size is determined by the mission requirements and cannot be arbitrarily reduced.

Background contrast is reduced primarily with camouflage paints, using colors and surface textures that cause the aircraft to reflect light at an intensity and color equal to that of the background. This requires assumptions as to the appropriate background as well as the lighting conditions.

Frequently aircraft will have a lighter paint on the bottom, because the background for look-up angles is the sky. Current camouflage paint schemes are dirty blue-grey for sky backgrounds and dull, mottled grey-greens and grey-browns for ground backgrounds.

Different parts of the aircraft can contrast against each other, which increases detectability. To counter this, paint colors can be varied to lighten the dark areas, such as where one part of the aircraft casts a shadow on another. Also, small lights can sometimes be used to fill in a shadow spot.

Canopy glint is also a problem for visual detection. The use of flat transparencies can be applied as previously discussed, but will tend to detract from the pilot's outside viewing.

At night, aircraft are visually detected mostly by engine and exhaust glow and by glint off the transparencies. These can be reduced by techniques previously discussed for IR and glint suppression.

There are also psychological aspects to visual detection. If the aircraft does not look like an aircraft, the human mind may ignore it. The irregular mottled patterns used for camouflage paints exploit this tendency.

In air-to-air combat, seconds are precious. If a pilot is confused as to the opponent's orientation, the opponent may obtain favorable positioning. To this end, some aircraft have even had fake canopies painted on the underside. Forward-swept and oblique wings may also provide momentary disorientation.

8.7 AURAL SIGNATURE

Aural signature (noise) is important for civilian as well as military aircraft. Commercial airports frequently have antinoise ordinances that restrict some aircraft. Aircraft noise is largely caused by airflow shear layers, primarily due to the engine exhaust.

A small-diameter, high-velocity jet exhaust produces the greatest noise, while a large-diameter propeller with a low tip-speed produces the least noise. A turbofan falls somewhere in between. Blade shaping and internal duct shaping can somewhat reduce noise.

Piston exhaust stacks are also a source of noise. This noise can be controlled with mufflers, and by aiming the exhaust stacks away from the ground and possibly over the wings.

Within the aircraft, noise is primarily caused by the engines. Well-designed engine mounts, mufflers, and insulation materials can be used to reduce the noise. Internal noise will be created if the exhaust from a piston engine impinges upon any part of the aircraft, especially the cabin.

Wing-mounted propellers can have a tremendous effect on internal noise. All propellers should have a minimum clearance to the fuselage of about 1 ft, and should preferably have a minimum clearance of about one-half of the propeller radius.

However, the greater the propeller clearance, the larger the vertical tail must be to counter the engine-out yaw.

Jet engines mounted on the aft fuselage (DC-9, B727, etc.) should be located as far away from the fuselage as structurally permitted to reduce cabin noise.

8.8 VULNERABILITY CONSIDERATIONS

Vulnerability concerns the ability of the aircraft to sustain battle damage, continue flying, and return to base. An aircraft can be “killed” in many ways. A single bullet through a non-redundant elevator actuator is as bad as a big missile up the tailpipe!

“Vulnerable area” is a key concept. This refers to the product of the projected area (square feet or meters) of the aircraft components, times the probability that each component will, if struck, cause the aircraft to be lost. Vulnerable area is different for each threat direction. Typical components with a high aircraft kill probability (near 1.0) are the crew compartment, engine (if single-engined), fuel tanks (unless self-sealing), and weapons. Figure 8.14 shows a typical vulnerable area calculation.

When assessing the vulnerability of an aircraft, the first step is to determine the ways in which it can be “killed.” Referred to as a “failure modes and effects analysis (FMEA),” this step will typically be performed during the later stages of conceptual design. The FMEA considers both the ways in which battle damage can affect individual aircraft components, and the ways in which damage to each component will affect the other components.

During initial configuration layout, the designer should strive to avoid certain features known to cause vulnerability problems. Fire is the greatest danger to a battle-damaged aircraft. Not only is the fuel highly flammable, but so is the hydraulic fluid. Also, combat aircraft carry gun ammo, bombs, and missiles. An aircraft may survive a burst of cannon shells only to explode from a fire in the ammo box.

If at all possible, fuel should not be located over or around the engines and inlet ducts. While tanks can be made self-sealing to a small puncture, a large hole will allow fuel to ignite on the hot engine. The pylon-mounted engines on the A-10 insure that leaking fuel cannot ignite on the engines.

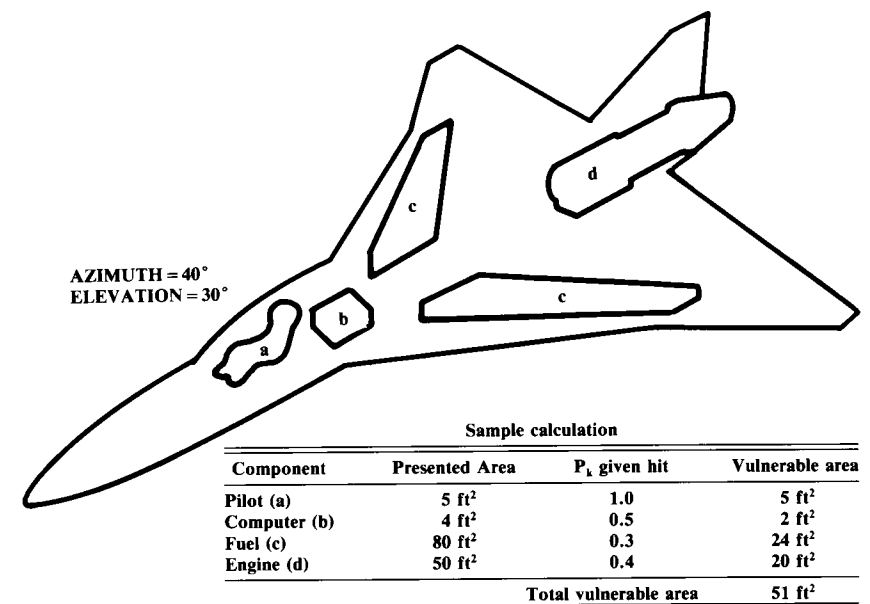


Fig. 8.14 Vulnerable area calculation.

Similarly, hydraulic lines and reservoirs should be located away from the engines.

Firewalls should be used to prevent the spread of flames beyond a burning engine bay. Engine bays, fuel bays, and weapon bays should have a fire-suppression system.

When an engine is struck, turbine and compressor blades can fly off at high speeds. Avoid placing critical components such as hydraulic lines or weapons anywhere where they could be damaged by an exploding engine. Also, a twin-engine aircraft should have enough separation between engines to prevent damage to the good engine. If twin engines are together in the fuselage, a combined firewall and containment shield should separate them. This requires at least 1 foot of clearance between engines.

Propeller blades can fly off either from battle damage or during a wheels-up landing. Critical components, especially the crew and passenger compartments, shouldn't be placed within a 5-deg arc of the propeller disk.

Avoid placing guns, bombs, or fuel near the crew compartment. Fuel should not be placed in the fuselage of a passenger plane.

Redundancy of critical components can be used to allow the survival of the aircraft when a critical component is hit. Typical components that could be redundant include the hydraulic system, electrical system, flight control system, and fuel system. Note that while redundancy improves the survivability and reliability, it worsens the maintenance requirements because there are more components to fail.

For more information on vulnerability, Ref. 18 is again suggested.

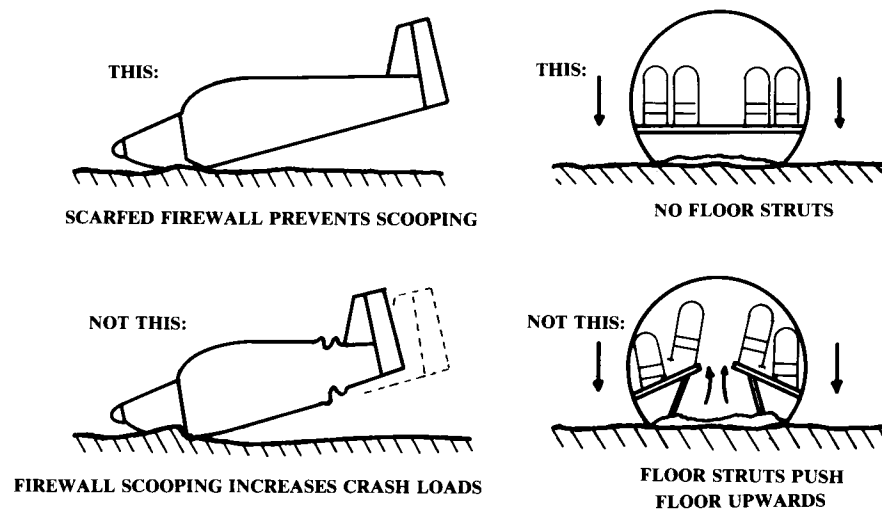


Fig. 8.15 Crashworthiness design.

8.9 CRASHWORTHINESS CONSIDERATIONS

Airplanes crash. Careful design can reduce the probability of injury in a moderate crash. Several suggestions have been mentioned above, including positioning the propellers so that the blades will not strike anyone if they fly off during a crash. Also mentioned was the desire to avoid placing fuel tanks in the fuselage of a passenger airplane (although fuel in the wing box carrythrough structure is usually acceptable).

Figure 8.15 shows several other design suggestions which were learned the hard way. A normal, vertical firewall in a propeller aircraft has a sharp lower corner which tends to dig into the ground, stopping the aircraft dangerously fast. Sloping the lower part of the firewall back as shown will prevent digging in, therefore reducing the deceleration.

For a large passenger aircraft, the floor should not be supported by braces from the lower part of the fuselage. As shown, these braces may push upward through the floor in the event of a crash.

Common sense will avoid many crashworthiness problems. For example, things will break loose and fly forward during a crash. Therefore, don't put heavy items behind and/or above people. This sounds obvious, but there are some aircraft with the engine in a pod above and behind the cockpit.

There are also some military jets with large fuel tanks directly behind the cockpit, offering the opportunity to be bathed in jet fuel during a crash. However, the pilot would probably try to eject rather than ride out a crash bad enough to rupture the fuel tanks.

One should also consider secondary damage. For example, landing gear and engine nacelles will frequently be ripped away during a crash. If possible, they should be located so that they do not rip open fuel tanks in the process.

Some form of protection should be provided in the not-unlikely event that the aircraft flips over during a crash. This is lacking in several small homebuilt designs.

8.10 PRODUCIBILITY CONSIDERATIONS

It is often said that aircraft are bought "by the pound." While it is true that aircraft cost is most directly related to weight, there is also a strong cost impact due to the materials selected, the fabrication processes and tooling required (forging, stamping, molding, etc.), and the assembly manhours.

The configuration designer does not usually determine the materials used or exactly how the aircraft will be fabricated. However, the ease of producing the aircraft can be greatly facilitated by the overall design layout.

The greatest impact the configuration designer has upon producibility is the extent to which flat-wrap structure is incorporated. This has a major impact upon the tooling costs and fabrication manhours, as discussed in the last chapter.

Part commonality can also reduce production costs. If possible, the left and right main landing gear should be identical (left-right common). It may be desirable to use uncambered horizontal tails to allow left-right commonality even if a slight aerodynamic penalty results. In some cases the wing airfoil can be slightly reshaped to allow left-right common ailerons.

Forgings are the most expensive type of structure in common usage, and are also usually the longest-lead-time items for production tooling. Forgings may be required whenever a high load passes through a small area. Forgings are used for landing-gear struts, wing-sweep pivots, and all-moving tail pivots (trunnions). The designer should avoid, if possible, such highly-loaded structure.

Installation of internal components and routing of hydraulic lines, electrical wiring, and cooling ducts comprises another major production cost due to the large amount of manual labor required. To ease installation of components and routing, avoid the tight internal packaging so desirable for reduced wetted area and wave drag. When evaluating proposed designs, government design boards will compare the overall aircraft density (weight divided by volume) with historical data for similar aircraft to insure packaging realism.

Routing can be simplified through provision of a clearly defined "routing tunnel." This can be internal or, as shown in Fig. 8.16, an external and nonstructural fairing that typically runs along the spine of the aircraft. However, if all routing is concentrated in one area the aircraft vulnerability will be drastically worsened.

Routing can be reduced by careful placement of the internal components. For example, the avionics and the crew station will both require cooling air ("environmental control"). If the avionics, crew station, and environmental control system (ECS) can be located near to each other, the routing distances will be minimized.

Sometimes clever design can reduce routing. The Rutan Defiant, a "push-pull" twin-engined design, uses completely separate electrical systems for the front and rear engines, including separate batteries. This requires an extra battery, but a trade study determined that the extra battery weighs less

than the otherwise-required electrical cable, and eliminates the front-to-rear routing requirement.

Another factor for producibility concerns manufacturing breaks. Aircraft are built in subassemblies. Typically, a large aircraft will be built up from a cockpit, an aft-fuselage, and a number of mid-fuselage subassemblies. A small aircraft may be built from only two or three subassemblies.

It is important that the designer consider where the subassembly breaks will occur, and avoid placing components across the convenient break locations. Figure 8.17 shows a typical fighter with a fuselage production break located just aft of the cockpit. This is very common because the cockpit pressure vessel should not be broken for fabrication.

In the upper design, the nose wheel well is divided by the production break, which prevents fully assembling the nose-wheel linkages before the two subassemblies are connected. The lower illustration shows a better arrangement.

Design for producibility requires experience that no book can provide. A good understanding of structural design and fabrication and the basic principles of operation for the major subsystems provides the background for developing producible designs. The following material provides a brief introduction to aircraft fabrication.

While there have been tremendous advances in aircraft production in recent years, much of the modern factory would be recognizable to a manufacturing engineer from the Wright Brothers' days. Aircraft production, then and now, involves the application of the mechanical arts of machining, forming, finishing, joining, assembly, and testing.

Machining involves the removal of a carefully-controlled amount of material from a part, typically by the application of a cutting tool via relative motion between the part and the tool. The cutting tool is generally based upon the inclined wedge, and acts to peel away a thin shaving of the part (a drill bit can be seen as a set of inclined wedges positioned radially around an axis). The relative motion between tool and part can be rotational, as with the drill, lathe, and mill, or it can be translational, as with the broach and planer.

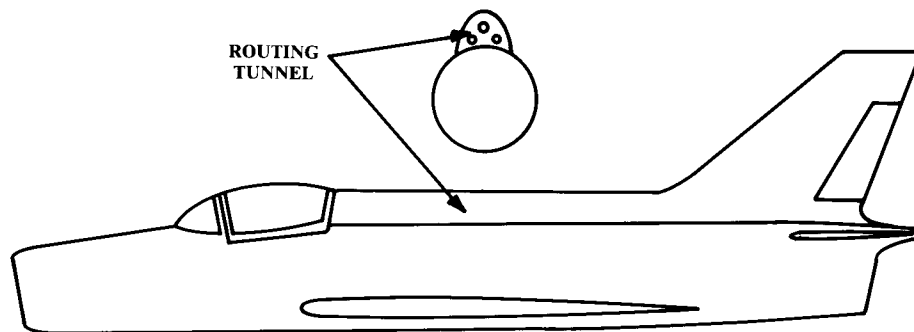


Fig. 8.16 External routing tunnel.

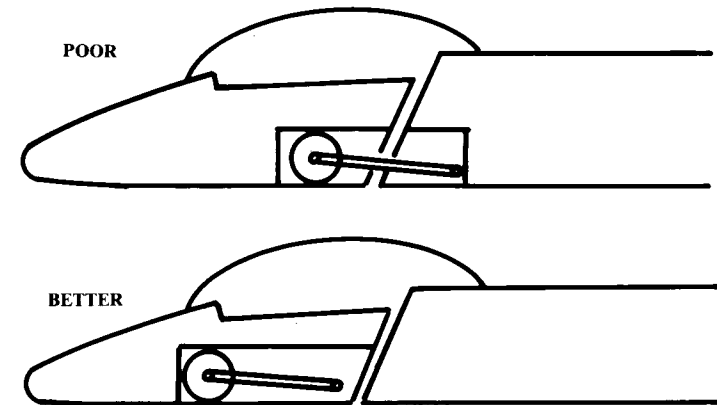


Fig. 8.17 Production breaks.

Forming refers to the numerous ways in which materials, especially metals, are changed in shape other than by machining. Forming includes casting, forging, extruding, stamping, punching, bending, and drawing. In casting, the metal is brought up to its melting temperature then poured into a mold. Forging involves forcing nonmolten metal into a mold through pressure or impact. Extrusion is the process of forcing metal to flow out a hole with the desired cross-sectional area, creating shaped bar stock. Stamping and punching are used to cut out shapes and holes in sheet metal. Bending is self-explanatory, and drawing is the process of forcing sheet metal into a form creating cup-like geometries.

Finishing encompasses a number of processes applied to formed and/or machined parts. Some finish processes include further material removal to create a smoother surface, such as deburring, lapping, and finish grinding. Other finish processes, such as painting, anodization, and plating, involve application of a surface coating.

Composite fabrication is sufficiently unlike metal fabrication that it deserves special mention. In thermoset composite production, a liquid or pliable semisolid plastic material undergoes a chemical change into a new, solid material, usually accompanied by the application of heat and/or pressure. For aircraft applications the plastic "matrix" material is reinforced by a fiber, typically of graphite material. Thermoset composite manufacture is unique in that the material itself is produced at the same time and place as the part. A second class of composites, the thermoplastics, involves a plastic matrix which is heated in a mold until it deforms readily, assuming the shape of the mold. Composite fabrication is further described in Chapter 14.

Joining is simply the attachment of parts together, by processes including brazing, soldering, welding, bonding, riveting, and bolting. All these processes historically have a high manual-labor content, and all are being automated to various extents in modern factories. For example, modern car factories have long lines of robotic spot-welders attaching body panels.

Automatic riveting machines, applicable for simple geometries such as rivets in a row down a wing spar, can be found in the modern aircraft factory.

Assembly is the process of combining parts and subassemblies into the final product. Assembly usually involves joining operations such as riveting or bolting, but is distinguished from joining by the greater level of completeness of the subassemblies. For example, when you attach a wing skin to the wing ribs it is "joining," but when you attach the wing to the fuselage, it is "assembly."

Testing is a key part of the manufacturing process. In traditional factories, testing was generally done by random selection of finished product and was frequently of a destructive nature. While helping to keep average quality up, such random destructive testing did not insure that any given part was acceptable because the only parts known by testing to be acceptable were destroyed in the process!

Today's factories are tending toward nondestructive testing techniques such as magnaflux, ultrasonic, and nuclear magnetic resonance, and are also applying advanced statistical techniques to better select samples to test and to determine the corrective action required.

CAD/CAM, or Computer-Aided Design/Computer-Aided Manufacture, is a generic term for the many different ways in which computers are being used in design and manufacture. Typically CAD/CAM refers specifically to the use of computers for two- and three-dimensional component design, and the use of the resulting CAD data base as the input for the programming of numerically-controlled machinery and robots (as described below). The benefits of CAD/CAM are well-established and include improved design quality, reduction in design time and/or increase in the number of design iterations possible, earlier discovery of errors, integration of design, analysis, and manufacturing engineering, and facilitation of training.

Automation refers to almost any use of computerized equipment during manufacture. However, the generic term "automation" is most frequently applied to tasks such as riveting, parts retrieval, and process control (such as autoclave cycling), whereas the more specific terms "numerical control" and "robotics" are used as described below.

Numerical control (NC) programming refers to the creation of digital instructions which command a computer-controlled machine tool such as a mill or lathe. This area is probably one of the highest leverage in terms of reducing cost and improving quality. While machine tools themselves have experienced little fundamental change in this century (this author knows of a company making high-tech wind turbines on a 100-year-old lathe!), the application of numerical control replacing the skilled but bored machinist has had a tremendous effect on productivity and quality.

The most sophisticated subset of automation is robotics, in which a computer-controlled machine performs tasks involving highly complex motions which previously might have been performed by a human. Note that it is the ability to physically manipulate objects in response to programming which distinguishes the robot from other forms of automation or mechanism. Robotics examples include part pickup and positioning, painting,

composite-ply laydown, material handling, simple assembly, and welding, and are usually limited to "semi-skilled" jobs, at least to date.

A key robotics technology for composites is in the labor-intensive tape lay-up process. Programmable robot arms with tape dispenser end effectors are widely used to place the prepreg. Also, autoclave cycle control is widely automated.

Rapid prototyping of parts without tools is being performed using a new technique known as Stereolithography (SLA), which can produce plastic prototype parts in a day or less. SLA works by mathematically slicing CAD designs into thin cross sections, which are traced one at a time by an ultraviolet laser beam on a vat of photosensitive chemicals that solidify as they are irradiated. After each layer is completed, an "elevator" holding the part moves down slightly and the next layer is solidified on top of it. While to date only certain types of relatively fragile plastics may be used by SLA devices, the plastic prototypes can then be used to create molds for strong epoxy or aluminum parts.

8.11 MAINTAINABILITY CONSIDERATIONS

Maintainability means simply the ease with which the aircraft can be fixed. "Reliability and Maintainability" (R&M) are frequently bundled together and measured in "Maintenance Manhours Per Flighthour" (MMH/FH). MMH/FH's range from less than one for a small private aircraft to well over a hundred for a sophisticated supersonic bomber or interceptor.

Reliability is usually out of the hands of the conceptual designer. Reliability depends largely upon the detail design of the avionics, engines, and other subsystems. The configuration designer can only negatively impact reliability by placing delicate components, such as avionics, too near to vibration and heat sources such as the engines.

Anybody who has attempted to repair a car will already know what the major driver is for maintainability. Getting at the internal components frequently takes longer than fixing them! Accessibility depends upon the packaging density, number and location of doors, and number of components that must be removed to get at the broken component.

Packaging density has already been discussed. The number and location of doors on modern fighters have greatly improved over prior-generation designs. Frequently the ratio between the total area of the access doors and the total wetted area of the aircraft's fuselage is used as a measure of merit, with modern fighters approaching a value of one-half.

A structural weight penalty must be paid for such access. This leads to the temptation to use "structural doors" that carry skin loads via heavy hinges and latches. These are always more difficult to open than nonload-bearing doors because the airframe's deflection from its own weight will bind the latches and hinges. In extreme cases, the aircraft must be supported on jacks or a cradle to open these structural doors.

As a general rule, the best access should be provided to the components that break the most often or require the most routine maintenance. Engine access doors should definitely be provided that allow most of the engine to be exposed. Also, large doors should be provided for the avionics compart-

ment, hydraulic pumps, actuators, electrical generators, environmental control system, auxiliary power unit, and gun bay.

The worst feature an aircraft can have for maintainability is a requirement for major structural disassembly to access or remove a component. For example, the V/STOL AV-8B Harrier requires that the entire wing be removed before removing the engine. Several aircraft require removal of a part of the longeron to remove the wing.

Similarly, the designer should avoid placing internal components such that one must be removed to get to another. In the F-4 Phantom, an ejection seat must be removed to get to the radio (a high-break-rate item). It is not uncommon for the ejection seat to be damaged during this process. "One-deep" design will avoid such problems.

CREW STATION, PASSENGERS, AND PAYLOAD

9.1 INTRODUCTION

At the conceptual design level it is not necessary to go into the details of crew-station design, such as the actual design and location of controls and instruments, or the details of passenger and payload provisions. However, the basic geometry of the crew station and payload/passenger compartment must be considered so that the subsequent detailed cockpit design and payload integration efforts will not require revision of the overall aircraft.

This chapter presents dimensions and "rule-of-thumb" design guidance for conceptual layout of aircraft crew stations, passenger compartments, payload compartments, and weapons installations. Information for more detailed design efforts is contained in the various civilian and military specifications and in subsystem vendors' design data packages.

9.2 CREW STATION

The crew station will affect the conceptual design primarily in the vision requirements. Requirements for unobstructed outside vision for the pilot can determine both the location of the cockpit and the fuselage shape in the vicinity of the cockpit.

For example, the pilot must be able to see the runway while on final approach, so the nose of the aircraft must slope away from the pilot's eye at some specified angle. While this may produce greater drag than a more-streamlined nose, the need for safety overrides drag considerations. Similarly, the need for over-side vision may prevent locating the cockpit directly above the wing.

When laying out an aircraft's cockpit, it is first necessary to decide what range of pilot sizes to accommodate. For most military aircraft, the design requirements include accommodation of the 5th to the 95th percentile of male pilots, (i.e., a pilot height range of 65.2–73.1 in.). Due to the expense of designing aircraft that will accommodate smaller or larger pilots, the services exclude such people from pilot training.

Women are only now entering the military flying profession in substantial numbers, and a standard percentile range for the accommodation of female pilots had not yet been established as this was written. Future military aircraft might require the accommodation of approximately the 20th percentile female and larger. This may affect the detailed layout of cockpit controls and displays, but should have little impact upon conceptual cockpit layout.